Name: ___________________________________ Class: _________ Index: ___________

CATHOLIC JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION

BIOLOGY
9747/01
Paper 1

Wednesday 12 September 2008

1 hour 15 min

Additional materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or 2B recommended)

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.
Write and/or shade your name, NRIC / FIN number and HT group on the answer sheet in the spaces provided.

There are **40** questions in this paper. Answer **all** questions. For each question, there are four possible answers, **A, B, C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.

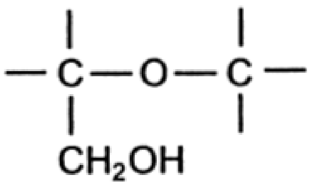
Calculators may be used

This document consists of **17** printed pages.
[Turn over]

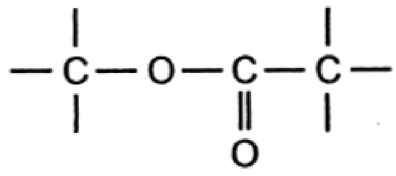
Section A

1 Which set of bonds is found in triglycerides?

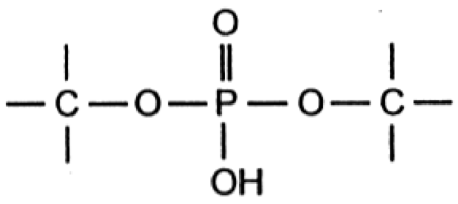
A



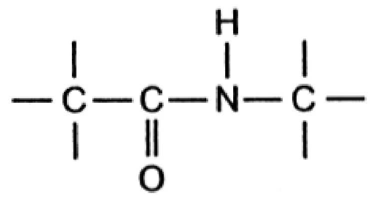
B



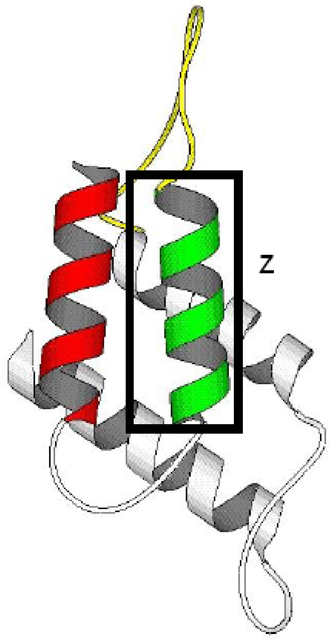
C



D



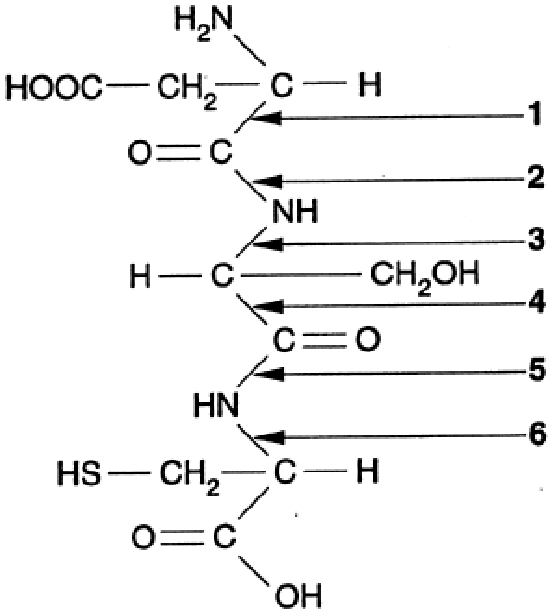
2 The diagram below shows the structure of a toxin found in cobra venom.



Which correctly describes the structure of the part enclosed in box Z?

	Number of Amino Acids Present	Types of bonds Present
A	Cannot be determined	Hydrogen Bonds only
B	Cannot be determined	Hydrogen Bonds and hydrophilic bonds
C	Approximately 11 amino acids	Hydrogen bonds only
D	Approximately 11 amino acids	Hydrogen bonds and hydrophobic bonds

3 The diagram represents a tripeptide.



At which bonds does hydrolysis occur to release the amino acids from the tripeptide?

A 1 and 4

B 1 and 6

C 2 and 5

D 3 and 6

4 What is the function of the different types of sugar units that are attached to the glycoproteins of cell surface membranes?

A to enable the molecule to act as an ion channel

B to ensure that the glycoprotein remains attached to phospholipids

C to give the molecule a specific shape enabling pinocytosis to occur

D to give the molecule a specific shape enabling the cell to interact with other cells

5 Each polypeptide chain of haemoglobin contains a number of regions in the form of an alpha helix.

Which feature of the haemoglobin molecule is responsible for this?

A bonding between the four polypeptide chains

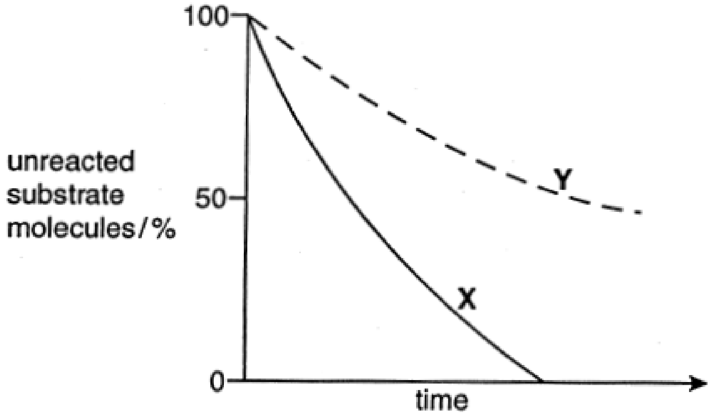
B hydrophobic interactions at the centre of the molecule

C the amino acid sequence of the globin

D the presence of iron in the haem groups

6 Curve **X** represents the course of an enzyme-catalysed reaction under optimum conditions.

Curve **Y** shows the action of the same enzyme on the same substrate but with one alteration to the reaction conditions.



Which factor, operating to a constant extent throughout the experiment, could give the results shown by curve **Y**?

A increased substrate concentration

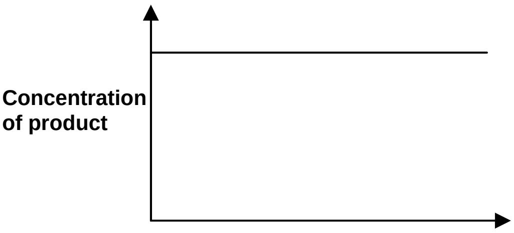
B inhibition by the end product

C less concentrated reaction mixture

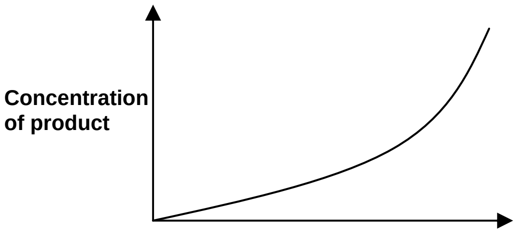
D lower temperature

7 Which graph shows the effect of increasing enzyme concentration, when both temperature and substrate concentration is limiting, and amount of time is fixed and in excess?

A



B



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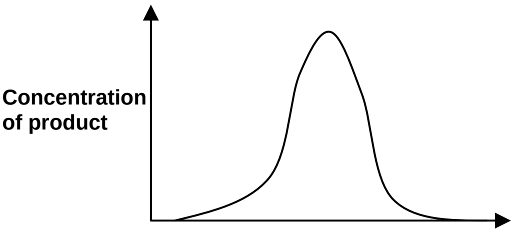

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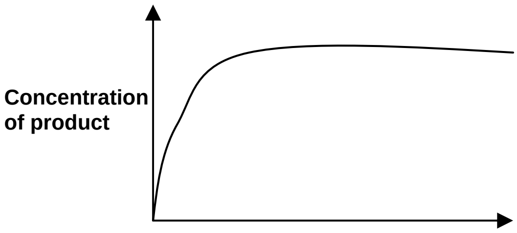


8 Liver cells were homogenized and their organelles separated in a centrifuge. Three main types of organelles were identified when the separation was completed.

1 large, mainly spherical organelles, easily separated out intact

2 smaller, elongated organelles with characteristic folded inner membranes

3 very small particles present in very large numbers

Which one of the following (A to D) best describes the organelles obtained?

	Contain DNA	Site of Krebs cycle	Site of protein synthesis
A	3	2	1
B	1	2	3
C	2	1	3
D	1	3	2

9 The amount of DNA in a mammalian cell in early prophase I of meiosis is **X**.

What is the amount of DNA in the same cell at G1 of Interphase ?

A 0.25**X**

B 0.5**X**

C **X**

D 2**X**

10 The second division of meiosis differs from mitosis because in meiosis

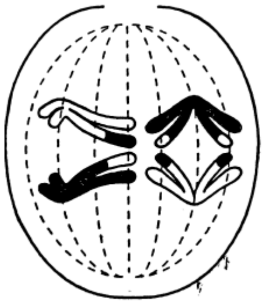
A chiasmata form between the chromatids

B each chromosome replicates at metaphase

C individual chromosomes line up at random on the equator

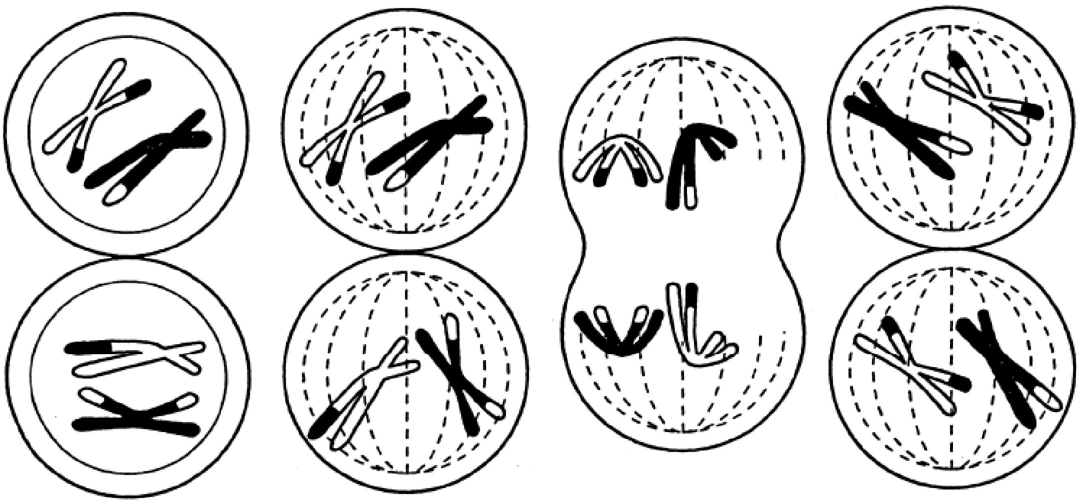
D the separating chromatids differ genetically

11 The diagram shows anaphase I of meiosis.



Which diagram shows metaphase II as meiosis continues in this cell?

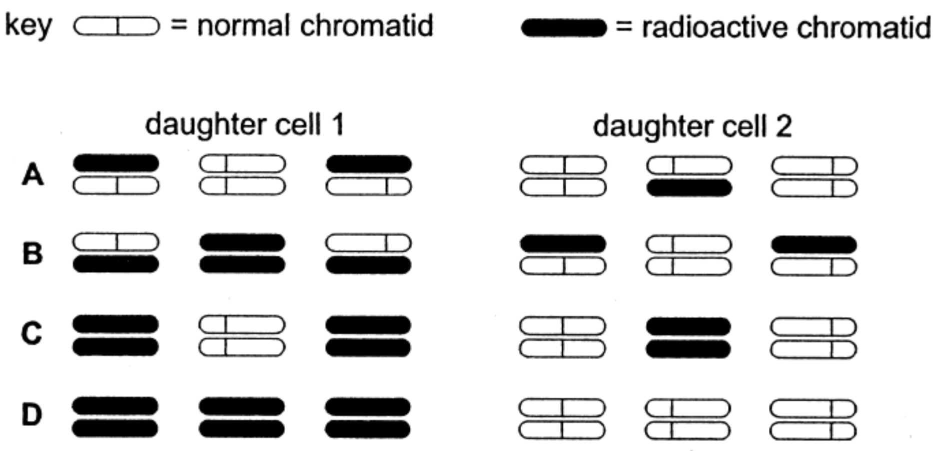
A **B** **C** **D**



12 The following experiment was carried out.

1. Haploid cells, containing three chromosomes each, were grown in a medium containing radioactive thymine, so that all the DNA was labelled.
2. Cells in early interphase were then transferred to a medium where the available thymine was not radioactive.
3. A single cell was immediately isolated and allowed to divide once. When the two daughter cells reached the next metaphase, they were fixed and their three chromosomes were inspected for radioactivity.

Which diagram represents the distribution of radioactivity at metaphase in the two daughter cells?



13 One complete turn of the double helix of DNA contains 10 pairs of bases and is 3.4 nm long.

What is the approximate length of the DNA coding sequence of lysozyme, a protein of 129 amino acids?

A 132 nm

B 113 nm

C 66 nm

D 44 nm

14 Insulin is a protein containing 51 amino acids. These include 17 of the 20 different amino acids commonly occurring in proteins.

What is the minimum number of different kinds of tRNA molecules involved in the synthesis of insulin?

A 3

B 17

C 20

D 51

15 Part of the amino acid sequences in normal and sickle cell haemoglobin are shown.

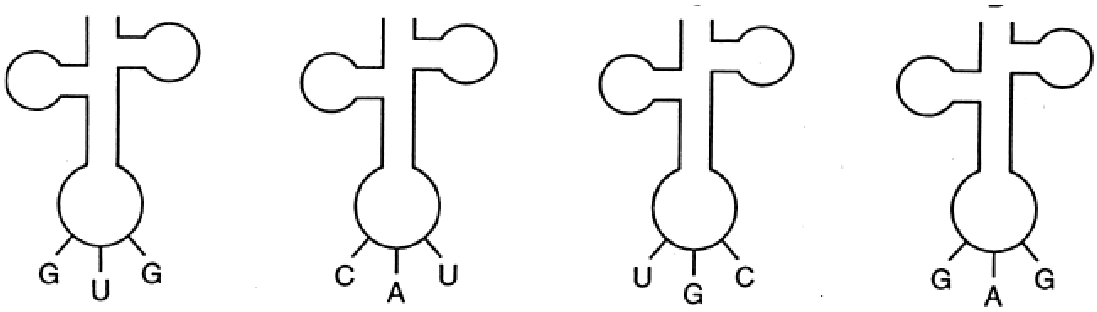
<u>normal haemoglobin</u>	<u>sickle cell haemoglobin</u>
thr-pro-glu-glu	thr-pro-val-glu

mRNA codons for these amino acids are:

glutamine (glu) GAA GAG
proline (pro) CCU CCC
threonine (thr) ACU ACC
valine (val) GUA GUG

Which transfer RNA molecule is involved in the formation of this part of the sickle cell haemoglobin?

A **B** **C** **D**



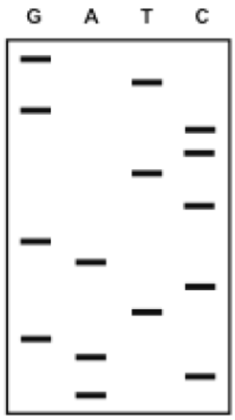
16 The table shows the results of an analysis of percentage concentration of three bases in nucleic acids from four sources. Three of the sources are DNA and one is RNA.

Which source is RNA?

source	adenine	cytosine	guanine
A	18.9	31.9	30.2
B	25.5	24.6	23.8
C	26.7	28.8	22.3
D	31.1	18.3	18.7

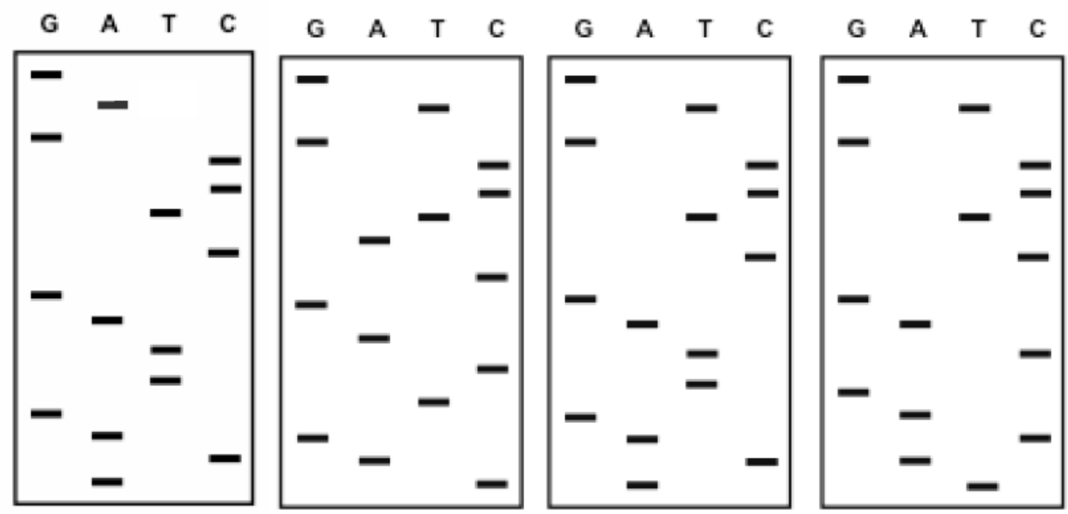
17 In people with a gene mutation where a base insertion has occurred, the protein formed has a very different primary structure. To identify the presence of this mutant allele, DNA nucleotide sequences were compared using gel electrophoresis.

The electrophoresis results from the DNA of a normal allele of a gene is shown below.



Which diagram represents the DNA sequence for the mutant allele of this gene?

A **B** **C** **D**



18 To what extent can glucose be oxidised by the soluble portion of cytoplasm of muscle cells, from which all organelles have been removed?

A Acetyl-CoA but no further

B Carbon dioxide and water

C Lactate but no further

D Pyruvate but no further

19 Dinitrophenol is a metabolic poison that can lodge within the thylakoid membranes of chloroplasts. It then provides an alternate routes for H^+ ions to diffuse across the thylakoid membranes. In what way will the Calvin cycle (dark reaction) be affected in chloroplasts poisoned with dinitrophenol?

A No effect since Calvin cycle is an enzyme –controlled process.

B The rate of Calvin cycle will increase because pH in the stroma will drop towards the optimum.

C The rate of Calvin cycle will decrease with the accumulation of glycerate-3-phosphate (GP)

D The rate of Calvin Cycle will decrease with the accumulation of glyceraldehydes-3-phosphate i.e. Triose Phosphate (TP).

20 The weed killer DCMU blocks the flow of electrons from the electron transport chains in photophosphorylation.

Why does this kill the plant?

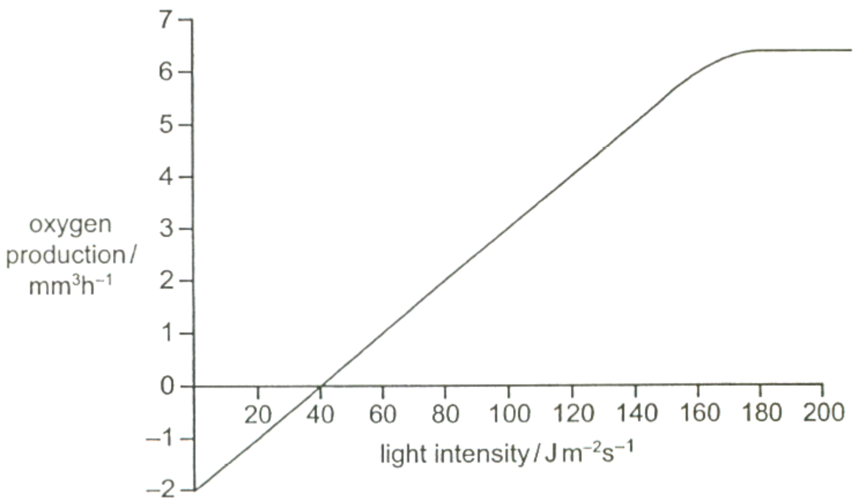
A Active transport of mineral ions is prevented.

B ATP and reduced NADP are not produced.

C Photoactivation of the chlorophyll cannot occur.

D Photolysis of water does not occur.

21 The graph shows the relationship between oxygen production in photosynthesis and light intensity for a unicellular green organism in 0.02% sodium hydrogencarbonate solution.



What is the most likely explanation for the graph leveling off at 180 J $m^{-2}s^{-1}$?

A light limited and carbon dioxide saturated

B light limited and the temperature is below optimum

C light saturated and carbon dioxide limited

D light saturated and the temperature is above optimum

22 The table shows events that may occur in the Calvin cycle, the Krebs cycle or in glycolysis.

In which of these processes do the events occur?

	ATP used	dehydrogenation	reduced NADP used
A	Calvin cycle	glycolysis	Krebs cycle
B	glycolysis	Calvin cycle	Krebs cycle
C	glycolysis	Krebs cycle	Calvin cycle
D	Krebs cycle	glycolysis	Calvin cycle

23 The drug atropine binds antagonistically to acetylcholine receptors. Nerve gases inhibit the activity of cholinesterase.

What will be the effects of these substances on the rate of transmission of impulses across a synapse?

	atropine	nerve gas
A	increased	increased
B	increased	reduced
C	reduced	increased
D	reduced	reduced

24 Dinitrophenol is a metabolic poison that prevents the production of ATP.

What will happen if a resting axon is treated with such a poison?

A The charge inside the axon membrane becomes more negative.

B The charge outside the axon membrane becomes less positive.

C The charge inside the axon membrane becomes less positive.

D The charge outside the axon membrane does not change.

25 In a small mammal, the allele for grey fur, **G**, is dominant to that for white fur, **g**. The allele for long tail, **T**, is dominant to the allele for short tail, **t**. Animals with grey fur and long tails were crossed with those having white fur and short tails.

The table shows the phenotypes of the 55 offspring.

number of offspring	fur	tail
15	grey	long
14	grey	short
14	white	long
12	white	short

What were the genotypes of the parents?

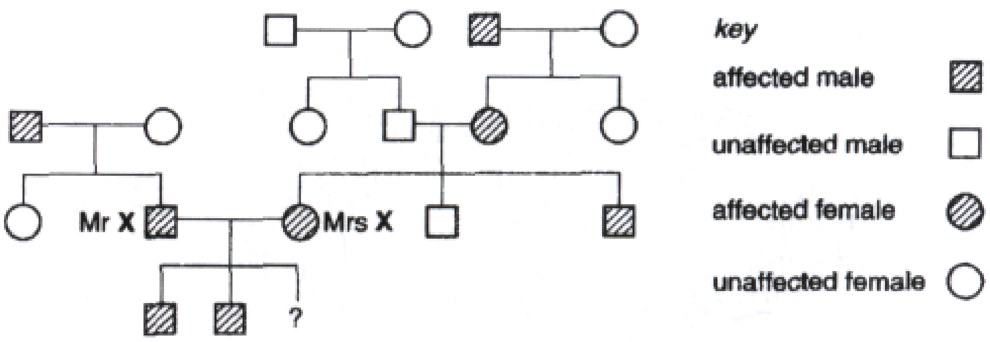
A GgTt x ggtt

B GGTt x Ggtt

C GgTT x Ggtt

D GgTT x ggtt

26 The family tree was assembled by a genetic counsellor for Mr and Mrs **X** who suffer from Marfan's Syndrome, resulting in serious defects in heart valves and lungs. Children who inherit the dominant mutant allele from both parents rarely survive beyond puberty without proper management.



What is the probability that Mr and Mrs **X**'s third child will be affected?

A 0.75

B 0.50

C 0.25

D 0.00

27 In guinea pigs, the allele **R** for rough coat is dominant over the allele **r** for smooth coat and the allele **B** for black fur is dominant over the allele **b** for white fur. The genes for fur colour and texture are not linked.

Two guinea pigs with genotype **RrBb** were mated together and one of the offspring had a rough, black coat.

What is the probability that this offspring was homozygous for both rough coat and black fur?

A 1 in 3

B 1 in 9

C 2 in 3

D 2 in 9

28 Two genes, Q and R affect the size of petals of a flower.

Gene Q has two alleles, Q^L and Q^A. The genotype $Q^L Q^L$ produces large petals, $Q^L Q^A$ produces small petals and $Q^A Q^A$ produces no petals.

Gene R has two alleles. R produces red pigment, and dominant over r that produces no pigment.

Two plants, heterozygous for both genes are crossed.

How many phenotypes are expected in the next generation?

A 4

B 6

C 9

D 12

29 The allele for an enzyme involved in the production of chlorophyll in geraniums is C^g. The mutant allele C^w codes for a defective gene with no pigment. A cross between two heterozygotes produced pale green plants and dark green plants in the ratio 2:1.

What is the most likely explanation for this ratio?

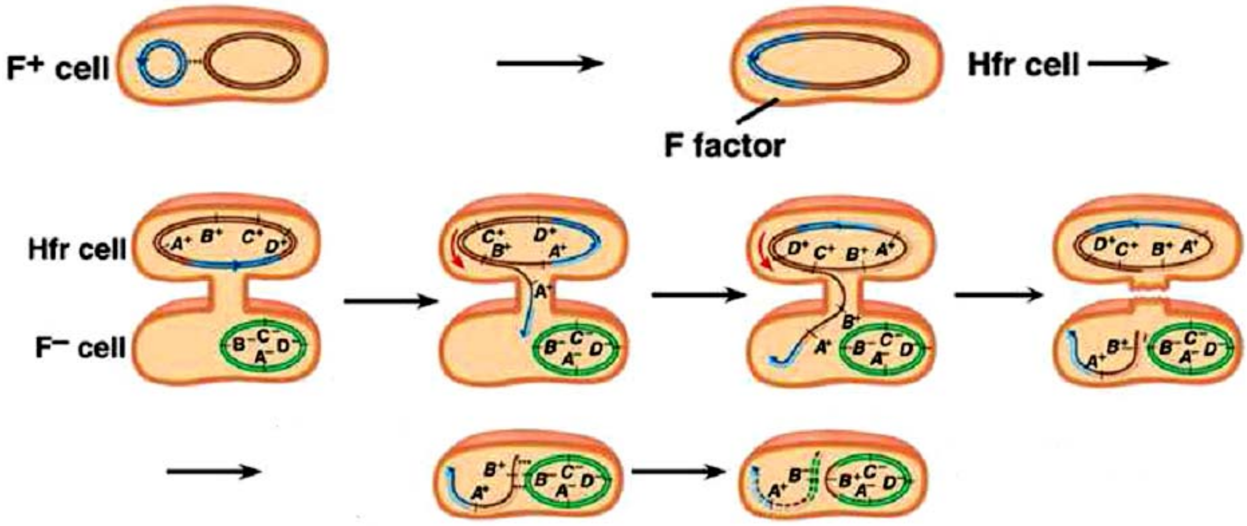
A C^g is a dominant allele.

B C^w is a dominant allele.

C $C^w C^g$ is a lethal phenotype.

D $C^w C^w$ is a lethal phenotype.

30 The diagrams below shows the process of conjugation in bacterial cells.



(i) Which genes will be replaced?
(ii) What is the process of replacing genes known as?
(iii) What type of recipient cells will result from the conjugation?

	Genes replaced	Process of replacing genes	Recipient cell after conjugation is completed
A	A, B	chiasmata	Partial diploid
B	B	crossing over	Partial diploid
C	A	chiasmata	F^+
D	B	crossing over	F^-

31 In Generalised Transduction, defective virus are formed as a result of;

A viral enzymes cutting the host DNA such that the host DNA is assembled into the new virus.

B production of host enzymes by virus which nicks its own DNA such that it can be assembled into the new virus.

C no shut down of host DNA production such that host DNA or the virus DNA can either be assembled into the new virus.

D integrate of virus DNA into host DNA and during excision the viral genome carries along with it the host DNA to be assembled into the new virus.

32 Which of the following viral structures are not completely coded by the virus' own nucleic acids?

A Capsid

B Glycoprotein gp140 of HIV

C Viral envelope

D Tail sheath of T4 bacteriophage

33 Four bacteriophage genes were mapped by recombination. The recombination frequencies between the four genes are shown in the table below.

	L	**M**	**N**	**P**
L	--	0.18	0.10	0.20
M		--	0.15	0.37
N			--	0.24
P				--

What is the order of the genes along the chromosome?

A L, M, N, P

B N, M, P, L

C M, L, P, N

D P, L, N, M

34 A mutation that renders the regulatory gene non-functional for the *lac* operon would result in;

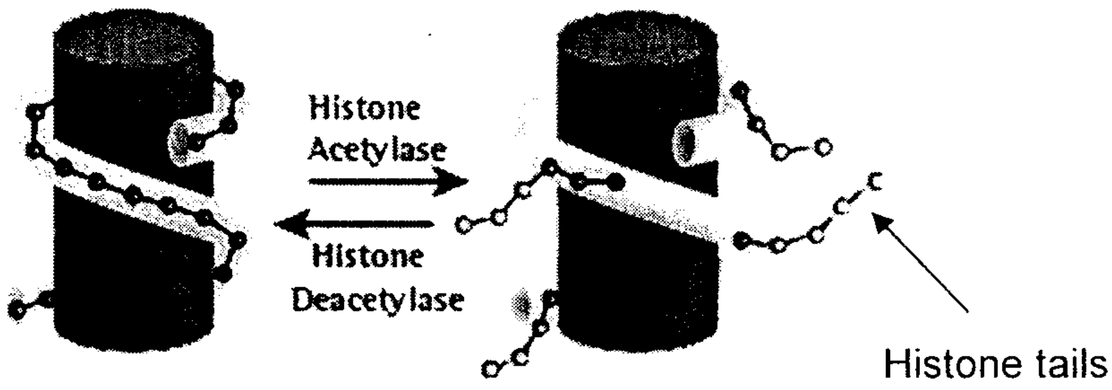
A Continuous production of beta galactosidase, regardless of concentration of lactose.

B Irreversible binding of repressor protein to the operator of *lac* operon.

C Complete blocking of the attachment site for RNA polymerase to the promoter.

D No effect on transcription rate, as operon is inducible.

35 As shown in the diagram below, acetylation of histones promotes loose chromatin structure. Recent evidence has shown that chemical modification of histones play a direct role in regulation of gene expression. Suggest how this works.



A Helicase action is enhanced by acetylation.

B Acetylation of histones neutralizes their negative charges and encourages binding to DNA polymerase.

C When nucleosomes are highly acetylated, chromatin becomes less compact and DNA is more accessible for transcription.

D RNA polymerase works better by binding with acetyl groups.

36 Which of the following increases the number of different alleles in a population?

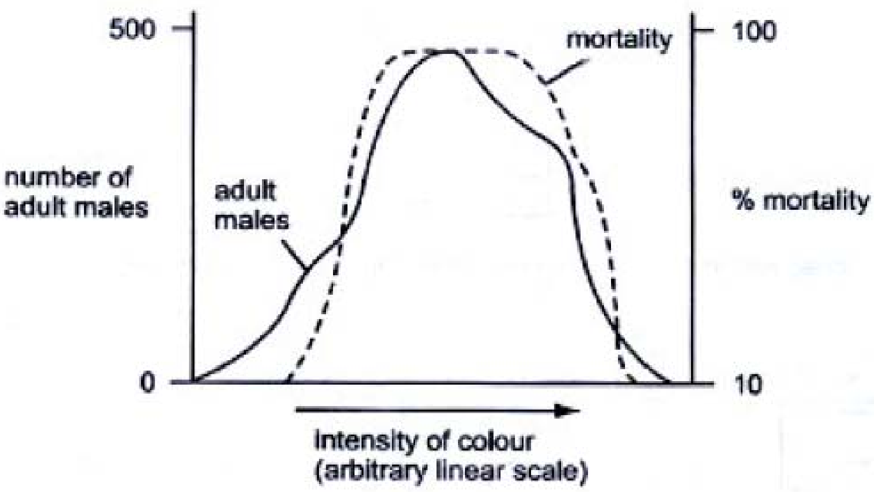
A crossing over

B gene mutation

C random fusion of gametes

D random assortment of chromosomes in meiosis

37 The graph below shows data on a population of a species of moth which shows considerable variation in colour intensity. Which conclusion can be made from this graph?



A Colour variation is environmentally induced.

B Colour variation is genetically determined.

C Extreme forms are favoured by natural selection.

D The species shows discontinuous variation with respect to colour.

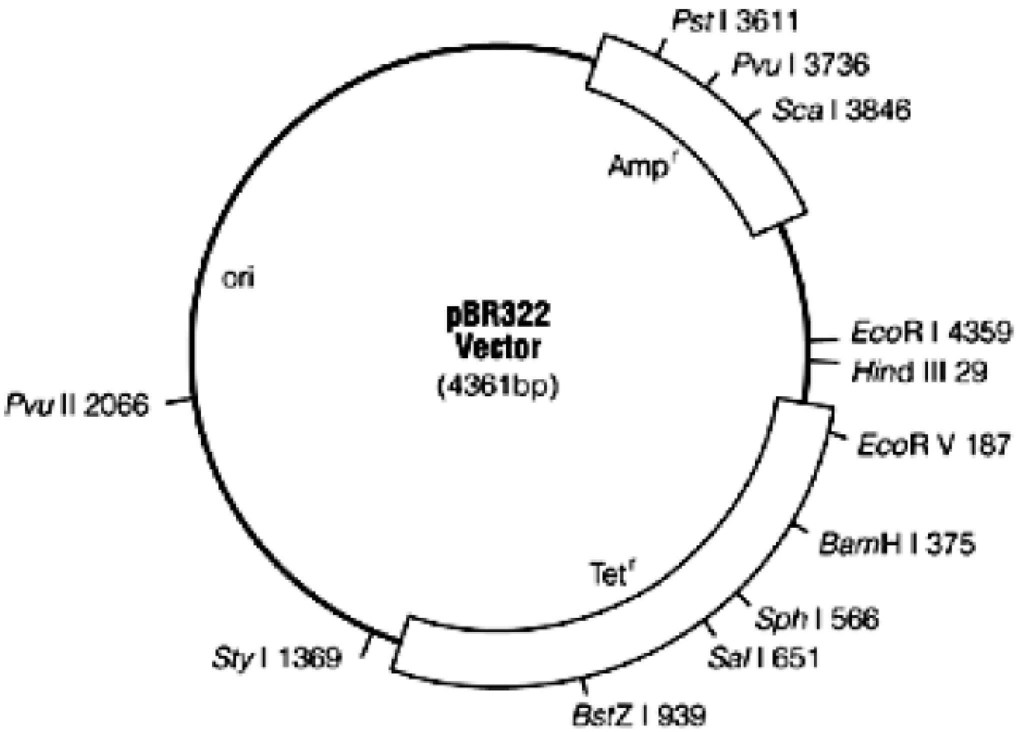
38 An insect collection contains 102 specimens of a species of butterfly. This specimen is sexually dimorphic, meaning that the males and females look different from each other. A student examined the specimens and collected the following data.

Observation	Frequency
Blue wing colour (male)	64
Brown wing colour (female)	38
Wing span 35-37 mm	15
Wing span 37-39 mm	68
Wing span 39-41 mm	19

How should this variation be classified?

	Continuous variation	Discontinuous variation
A	Sexual dimorphism	Colour
B	Colour	Sexual dimorphism
C	Sexual dimorphism	Wingspan
D	Wingspan	Sexual dimorphism

39 The pBR322 vector is used to clone a eukaryotic gene which has been digested by the restriction endonuclease BamHI.



Following transformation, bacterial cells were grown in four different media, as shown below:

I nutrient broth plus ampicillin

II nutrient broth plus tetracycline

III nutrient broth plus ampicillin and tetracycline

IV nutrient broth without antibiotics

Which of the following media would bacterial cells that contain the recombinant plasmids grow in?

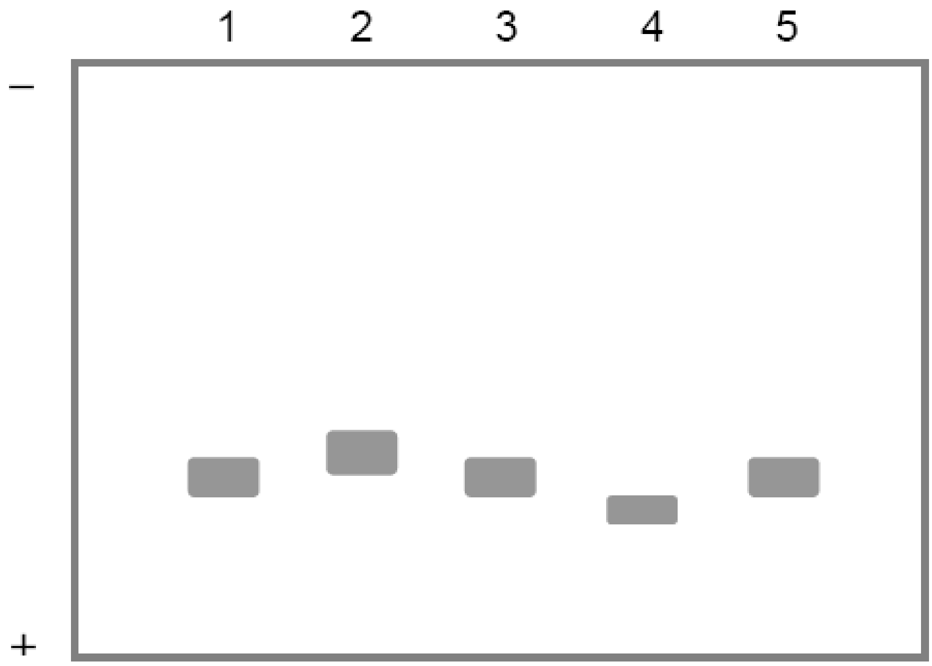
A I and II

B I and III

C I and IV

D IV only

40 The diagram shows the results of native polyacrylamide gel electrophoresis of the same protein extracted from five different individuals.



Which of the following correctly explains why the proteins extracted from individuals 2 and 4 did not travel the same distance on the gel as those extracted from individuals 1, 3 and 5?

	individual 2	individual 4
A	The protein has an additional negatively charged amino acid.	A non-charged amino acid in the protein has been replaced by a negatively charged amino acid.
B	The protein has an additional positively charged amino acid.	A non-charged amino acid in the protein has been replaced by a negatively charged amino acid.
C	The protein has an additional negatively charged amino acid.	A non-charged amino acid in the protein has been replaced by a positively charged amino acid.
D	The protein has an additional positively charged amino acid.	A non-charged amino acid in the protein has been replaced by a positively charged amino acid.